H100 queue profile comparison — same bins and request-size bands

≤704 own uncached · 705–4K own uncached · 4K–16K own uncached · 16K–48K own uncached · ≥48K own uncached

Offline: original profiling definition
Wait before + between prefill steps
Excludes all of the request's prefill-step time

Mean queue time per token-load bin (seconds)

Reconstructed remaining engine prefill tokens at arrival
4,456 requests in the displayed 0–100K range

Offline: first-scheduling queue time
Queue admission → first scheduled

Reconstructed remaining engine prefill tokens at arrival
4,456 requests in the displayed 0–100K range

Online: first-scheduling queue time
Queue admission → first scheduled

Router-estimated outstanding prefill tokens at arrival
312 requests in the displayed 0–100K range

0–30K zoom · Offline: original profiling definition
Wait before + between prefill steps
Excludes all of the request's prefill-step time

Mean queue time per token-load bin (seconds)

Reconstructed remaining engine prefill tokens at arrival
3,141 requests in the displayed 0–30K range

0–30K zoom · Offline: first-scheduling queue time
Queue admission → first scheduled

Reconstructed remaining engine prefill tokens at arrival
3,141 requests in the displayed 0–30K range

0–30K zoom · Online: first-scheduling queue time
Queue admission → first scheduled

Router-estimated outstanding prefill tokens at arrival
282 requests in the displayed 0–30K range

Hollow markers: fewer than 3 requests in the cell. Dashed horizontal line: 1 second. No regression is used in these curves.
Offline: 5 mixed-workload runs, no CPU KV offload. Online: 16-session ramp/hold, native CPU KV offload (64 GiB).
Online size bands use post-completion reported prompt minus cached tokens; the online model's x feature remains the router reservation estimate.